

Exploring Radiance Field Content Generation for Virtual Reality

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ABSTRACT

Generating content for virtual reality takes effort and expanding how content can be generated has the potential to increase participation by a wider range of users well outside virtual reality researchers and programmers. This work explores some current content acquisition and generation techniques for replicating real world 3D scenes and objects to better understand the practical use and application of these techniques. The techniques explored focus on the recent emergence of radiance field-based methods and their potential to replace previous scene/object-generating photogrammetry techniques.

Index Terms: Human-centered computing—Visualization—Visualization application domains—Information visualization

1 INTRODUCTION

Generating content for use in virtual reality (VR) or augmented reality (AR) experiences whether they are for games, training simulations or educational experiences often requires using pre-generated digital assets acquired from various websites or asset stores, hiring skilled people with 3D modeling experience to create custom models, or creating them oneself using various 3D modeling tools and algorithms. While it is possible to acquire all of the 3D models and assets needed to construct a specific scene, it is not easy and quite often, the acquired 3D content may not be exactly what was needed to fulfill the set of requirements. It is even more challenging if the people wanting to create a specific virtual 3D experience are not programmers, artists or VR/AR researchers.

This exploratory work examines recent 3D reconstruction algorithms for VR content generation. One of the specific goals is to better understand how these techniques work and might be utilized to generate specific content for VR/AR by a broader audience, thus potentially increasing participation in these areas. This work is motivated by teachers in high schools or colleges having access to VR headsets (*e.g.* our library maintains a fleet of VR headsets for use with classes). We aim to find effective and straight forward ways to place content development and creation for these instructors' VR exercises and experiences in their own hands, rather than having them rely on piecing together apps and external content.

2 RELATED WORK

In VR and AR, photogrammetry can be used to generate realistic 3D models or scenes. This technique involves capturing key features and reference points based on multiple, overlapping photographs of an object taken from different positions, covering the entire object in the process. This procedure leads to generating corresponding points in a 3D space; producing a dense point cloud. These points are then reconstructed into a surface mesh, with final details added through texturing the mesh with photo data. Photogrammetry offers various ways to customize and tweak each step but requires a handful of images taken from different angles if a detailed dense reconstruction is expected as an outcome. Along with that, the quality of images

and lighting conditions play a significant role as well. Maintaining all of these, as well as manual tweaking can be cumbersome for non-professionals, as well as specialized professionals.

The recent emergence of radiance field-based scene and object reconstruction methods that include several works based on Neural Radiance Field (NeRF) and 3D Gaussian Splatting are already showing promising results. These techniques could potentially be implemented in several VR applications, changing the traditional scene and object construction methods, and increase participation in the creation of specific VR and AR content.

Radiance field methods, particularly NeRF (Neural Radiance Fields) have received a lot of attention after being introduced by Mildenhall et al. in 2020 [4]. NeRF models construct a volumetric ray-marching based representation generated with neural networks. These models predict a continuous three-dimensional neural scene representation generated from images taken from different directions, similar to photogrammetry. Another recent method uses a 3D Gaussian-based radiance field [3] to achieve high quality visual qualities achieved like NeRF in a reasonable time. This work generates the radiance field by manipulating 3D Gaussians.

NeRF operations can be denoted by: $F(x, y, z, \theta, \phi) \rightarrow (c, \sigma)$. NeRF takes a continuous 5D coordinate (x, y, z, θ, ϕ) as input, which represents different components and dimensions of 2D images. Here, x, y , and z represent the coordinates of a point in the 3D spatial scene. These are used for determining the 3D structure and positions of objects and surfaces in the scene. Theta (θ) and phi (ϕ), represent camera orientation of the point with respect to the scene. NeRF models leverage this information to reconstruct the radiance and opacity of each 3D point. The view synthesis procedure uses volumetric ray-marching inside a Multi-layer Perceptron (MLP) to generate the continuous 3D representation. The output is a RGB color (c) and the volume density at that point (σ).

The Gaussian Splatting algorithm [3] takes a different approach to synthesizing 3D scenes. From the input images from different angles, it initiates a continuous sparse point cloud using Structure-from-Motion points as input. The key difference with NeRF models here is that the point cloud is represented with 3D Gaussians (which denotes a position, covariance matrix, and opacity α). The color within the radiance field is represented by spherical harmonics. During the optimization procedure, 3D Gaussian properties of 3D position, α (opacity), anisotropic covariance, and spherical harmonic coefficients get updated over multiple iterations.

The NeRF models, due to using a MLP to generate the continuous volumetric function, require a lot of computational power, but can estimate radiance better. These methods can better estimate spaces that were not captured during the data-capturing procedure, and present it through the radiance field representation.

The state-of-the-art NeRF work presented by Barron et al. [1] takes an average of 48 hours on a single Nvidia A100 GPU. Follow-up work improved the training time by 24 times while achieving identical results [2]. However, 3D Gaussian Splatting [3] can achieve a similar level of accuracy within 35-45 minutes, using the same hardware.

3 PRACTICAL EXPERIENCE RECONSTRUCTING VR SCENES

Work presented by Xu et al. produces remarkable results in constructing high-fidelity VR spaces using NeRF [6]. The data capture process requires sophisticated camera capture rigs, composed of an

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Figure 1: Left most image: ground truth image from the video supplied to the Gaussian splat algorithm. Middle images: screen capture from the VR rendering with Unity. Right image: participant viewing the 3D lab reconstruction.

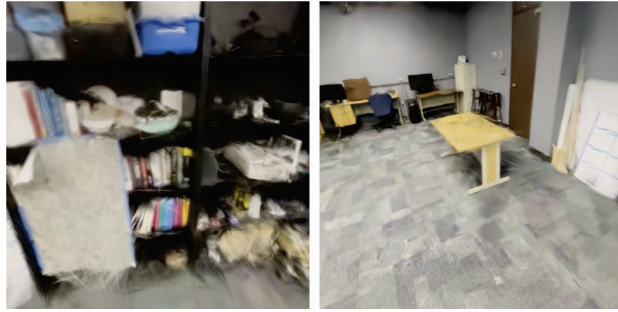


Figure 2: Acquisition and rendering errors from the Gaussian Splat radiance field. Objects away from the center of the room are subject to rendering distortions due to how the field is acquired.

array of 22 Sony $\alpha - 1$ mirrorless cameras. This solution may not readily be available to everyone wishing to create content in VR.

In this work, we explore the use of the 3D Gaussian Splatting model since it is faster to generate the resulting 3D dataset and the input for the algorithm can be acquired on devices that most people have, such as a mobile phone camera. We utilized an iPhone 12's Main (12MP and $f/1.6$ aperture) and Ultra Wide (12 MP, $f/2.4$ aperture and 120° field of view) camera to capture the assets for the representation model. We used an Oculus Rift S connected to a backpack PC and Unity to render the reconstructions in VR using the Unity Gaussian Splatting extension (<https://github.com/aras-p/UnityGaussianSplatting>). In this initial exploration, Gaussian splat processing was tested both with LumaAI ([lumalabs.ai](https://luma.ai)) as well as the official source of the Gaussian splat code. Further explorations will investigate specialized rendering and other alternatives for processing the Gaussian splat and NeRF-based models.

Figure 1 shows images from recreating a 21ft x 33ft lab space. The first capture is based on a 23-second video orbiting an object (a chair) generated by a radiance-field Gaussian splat representation. Following that, other captures involved similar attempts while taking longer and more stable 50-second recordings, with some captures that didn't strictly orbit around the object in the center of room.

The results were similar, in all the cases except the captures that did not orbit an object. Since the implemented structure-from-motion architecture requires multiple overlapping images captured from different angles to initiate the point cloud, the above-mentioned orbiting approach results in more accurate scene recreation. Thus, generating a reasonably accurate representation may be conveniently attainable.

4 DISCUSSION

Radiance field-based methods represent important methods for making content generation more exact, accessible and participatory, and the state-of-the-art radiance field studies show impressive results in generating realistic 3D scenes and objects [1]. Many examples focus All of this is very promising for content generation for VR

albeit with some current caveats. Since the NeRF-based methods represent the whole 3D representation through a fully connected neural network, producing radiance representation for all 3D points is computationally challenging and can take a much longer time to process, as compared with the 3D Gaussian Splat methods. Further, many current rasterization techniques are more optimized for mesh-based rendering, and future work will be needed to improve the rendering strategies for these types of models. In our exploration, the 3D-Gaussian-based representation leverages a fairly good result, especially in the center of the radiance field without a lot of processing time. This may make it more suitable, at least currently, for producing VR application assets.

There are limitations that should be noted. First, as seen in Figure 2, visual artifacts from the capture show sections of the reconstruction that are illegible (e.g. books on shelves), or rendered poorly. These artifacts are related to pixel resolution, sampling of the 3D space through the orbit technique, and distance from the center of the field. Some of these may be overcome by better processing of the input images, or by increasing the number of samples acquired. However, in our initial explorations, longer videos taken with slower movement through the capture space did not correlate with better 3D reconstructions. Additional improvement could also come from the rendering techniques applied on the headsets.

Another issue with radiance field approaches is that the representation is not explicit, unlike photogrammetry, modifying them or adding physics-based transformation is not possible at the moment. The lack of meshes makes these representations unable to implement interactive VR actions directly. Although there have been a few studies that came up with converting the radiance field to mesh [5], they still can not provide an accurate enough representation. Similarly, incorporating more than one radiance field representation into the same scene is challenging, although if possible, it could improve the scene representation even more through the capturing and blending of multiple objects into the same reconstructed environment.

5 CONCLUSION

This work presents an initial exploration using radiance field reconstruction for content generation in VR/AR and give initial ideas that the radiance field representation method can be an effective and intuitive way to generate realistic 3D content for these interactive environments. The results are impressive from fairly simple equipment. Our continued exploration of this topic involves determining ways in which the content generation can be improved and compared to other techniques. Further, addressing some of these presented limitations can make VR content generation more user-friendly, resulting in more collaborations on making and participating in VR applications.

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